

Binance BTCUSDT Historical Data Downloader & MultiCharts Converter

Overview

This project automates the complete process of downloading, extracting, validating, merging, and converting Binance BTCUSDT 1-minute historical market data into a format suitable for MultiCharts.

The script downloads all available monthly BTCUSDT Spot 1-minute archives from Binance Vision, extracts the raw CSV files, combines them into a single dataset, performs multiple validation and cleanup steps, resolves known data anomalies, and produces a final CSV file ready for import into MultiCharts.

The primary goal of the project is to eliminate the manual work normally required to maintain a complete BTCUSDT historical database while ensuring data consistency and compatibility with trading platforms.

Directory Structure

Default location:

```
C:\binance_data\BTCUSDT
|
├── zip
│   ├── BTCUSDT-1m-2017-08.zip
│   ├── BTCUSDT-1m-2017-09.zip
│   └── ...
├── csv
│   ├── BTCUSDT-1m-2017-08.csv
│   ├── BTCUSDT-1m-2017-09.csv
│   └── ...
└── BTCUSDT_1m_ALL.csv
```

The script can also be executed with a custom data directory supplied through the command line.

Example:

```
python binance_btcusdt_all.py "D:\MarketData\BTCUSDT"
```

Workflow

Step 1 - Generate Download List

The script automatically generates a list of monthly BTCUSDT 1-minute archives.

Start date:

2017-08

End date:

Last completed month

The current month is intentionally excluded because Binance may still be updating its contents.

Step 2 - Download Missing Archives

Each archive is downloaded from Binance Vision:

`https://data.binance.vision`

Before downloading, the script checks whether the file already exists.

If the archive is already present, it is skipped.

Benefits:

- Fast incremental updates
 - Reduced bandwidth usage
 - Safe re-execution of the script
-

Step 3 - Extract ZIP Archives

All downloaded archives are extracted into the CSV directory.

The script tracks:

- Total ZIP archives found
- Successfully extracted archives
- Skipped archives
- Extraction errors

Archives that have already been extracted are skipped automatically.

Step 4 - Load Historical Data

All extracted CSV files are loaded into memory.

For every file, the script attempts to read:

```
open_time
open
high
low
close
volume
```

Other Binance columns are ignored because they are not required for MultiCharts.

The script records:

- Total CSV files found
 - Successfully loaded files
 - Failed files
 - Total rows loaded
-

Step 5 - Timestamp Validation

Binance historical datasets are not always perfectly uniform.

The script validates all timestamps before processing.

Header Detection

Occasionally a CSV file may contain an unexpected header row.

The script converts timestamps to numeric values and removes invalid rows.

Reported as:

```
Header rows removed
```

Timestamp Range Validation

Only timestamps within the expected historical range are accepted.

Rows outside the valid range are removed.

Reported as:

Out-of-range rows removed

Microsecond Timestamp Detection

Some datasets may contain timestamps stored in microseconds instead of milliseconds.

Example:

1512367200000000

instead of

1512367200000

The script automatically detects and converts such values.

Reported as:

Microsecond timestamps detected

This prevents corrupted dates and duplicate bars caused by inconsistent timestamp precision.

Step 6 - Date and Time Conversion

After validation, Binance timestamps are converted into human-readable format:

MM/DD/YYYY
HH:MM

Example:

12/04/2017
06:00

This format is directly compatible with MultiCharts ASCII Import.

Step 7 - Duplicate Minute Bar Resolution

A known issue in some Binance historical datasets is the presence of multiple records that map to the same minute after conversion.

Example:

```
open_time=1512367200000  
Date=12/04/2017  
Time=06:00  
Volume=0.289485
```

```
open_time=1512367220799  
Date=12/04/2017  
Time=06:00  
Volume=0.000000
```

Although these timestamps are different, they both belong to the same minute.

MultiCharts cannot import duplicate minute bars.

To resolve this:

1. Duplicate Date+Time bars are detected.
2. Bars are sorted by volume.
3. The bar with the highest volume is preserved.
4. Lower-volume duplicates are removed.

This preserves the most meaningful market data while maintaining MultiCharts compatibility.

Step 8 - Export Final Dataset

The final dataset is written to:

```
BTCUSDT_1m_ALL.csv
```

Columns:

```
Date  
Time  
Open  
High  
Low  
Close  
Volume
```

Example:

```
12/04/2017,06:00,11476.87,11478.00,11476.87,11478.00,0.289485
```

The file is overwritten on every execution to ensure the latest history is always available.

Debug Logging

The script provides extensive diagnostic output during execution.

Download Phase

Reports:

```
Files to download
Archive already exists
Archive downloaded successfully
Archive not found
Download errors
```

Extraction Phase

Reports:

```
ZIP archives found
Archives extracted
Archives skipped
Extraction errors
```

Loading Phase

Reports:

```
CSV files found
CSV files loaded
CSV files failed
Rows loaded
```

Data Validation Phase

Reports:

```
Header rows removed
Microsecond timestamps detected
Out-of-range rows removed
Rows after cleanup
```

Duplicate Detection Phase

Reports:

```
Duplicate open_time values
Duplicate Date+Time values
Duplicate minute bars removed
```

If duplicates are found, detailed information is printed including:

```
Timestamp
Date
Time
Open
High
Low
Close
Volume
```

This information is useful for investigating data anomalies originating from Binance historical archives.

Export Phase

Reports:

```
Rows before cleanup
Rows saved
Historical range
Output file location
```

Example:

Rows saved: 4,321,500

Range:

08/17/2017 04:00

-> 05/31/2026 23:59

Output file:

C:\binance_data\BTCUSDT\BTCUSDT_1m_ALL.csv

Intended Use Cases

This project is designed for:

- MultiCharts users
- TradeStation users
- Quantitative traders
- Historical market data collection
- Strategy development
- Backtesting
- Research and analytics

It is particularly useful for traders who require a continuously updated BTCUSDT historical database without manually downloading and processing Binance archives.

Key Features

- Automatic Binance archive discovery
- Incremental downloading
- Automatic ZIP extraction
- Historical data merging
- Timestamp normalization
- Microsecond-to-millisecond correction
- Data validation
- Duplicate minute bar resolution
- MultiCharts-compatible export
- Detailed diagnostic logging
- Re-runnable without manual cleanup
- Custom data directory support
- Complete BTCUSDT 1-minute history generation with a single command